Problem 1 | Problems T6andT7_Electric and Magnetic Properties.pdf #1 | Resistivity · temperature coefficient

Silver and Aluminium Wires with the Same Resistivity

Original statement · Problems T6andT7_Electric and Magnetic Properties.pdf

1. At what temperature will a silver wire exhibit the same resistivity as an aluminium wire at 45 ºC. The resistivities of silver and aluminium at 0 ºC are $1.47 \cdot 10^{-6}$ *and* $2.70 \cdot 10^{-6}$ *Ω·cm respectively and their resistivity coefficients, 0.0038 and 0.0039 ºC⁻¹, respectively.*

Official Solution: 305 °C

HYPOTHESES & PARAMETERS

Hypotheses

1. **Matthiessen's rule.** The resistivity of a metal is the sum of a thermal part and a residual part, [Session 13 Slide 14]. The residual part (impurities and defects) is already contained in the resistivity measured at 0 °C, so each wire is described by one number ρ_0 and one coefficient α_T.
2. **Linear thermal resistivity.** The thermal part grows linearly with the temperature measured from 0 °C, with the temperature coefficient of resistivity α_T [Session 13 Slide 14]. The sheet gives one constant α_T per metal, so it is used over the whole range up to the result (an extrapolation that the sheet asks for; real silver departs slowly from a straight line).
3. **Same wire geometry is irrelevant.** Resistivity is a property of the material, not of the length or section, so the wires need no dimensions.

Data

Quantity	Symbol	Value as printed	In SI
Resistivity of Ag at 0 °C	$\rho_{0,\mathrm{Ag}}$	1.47×10^{-6} Ω·cm	1.47×10^{-8} Ω·m
Resistivity of Al at 0 °C	$\rho_{0,\mathrm{Al}}$	2.70×10^{-6} Ω·cm	2.70×10^{-8} Ω·m
Coefficient of Ag	α_{Ag}	0.0038 °C⁻¹	0.0038 K⁻¹
Coefficient of Al	α_{Al}	0.0039 °C⁻¹	0.0039 K⁻¹
Temperature of the Al wire	T_{Al}	45 °C	45 °C (a difference from 0 °C: 45 K)

Unknown: the temperature T_{Ag} at which silver has the resistivity of the aluminium wire at 45 °C.

The sheet prints the resistivities in Ω·cm and without the Ω and the dot in the first value; they are read from Slide 15, where the same problem is solved (ρ_0 in $\mu\Omega$·cm on Slide 14: Al 2.7, Ag 1.47).

MODEL & STRATEGY

Aluminium is more resistive than silver at 0 °C, so silver has to be heated until the growth of its thermal resistivity makes up the difference. The figure shows the two straight lines and the horizontal level that fixes the answer.

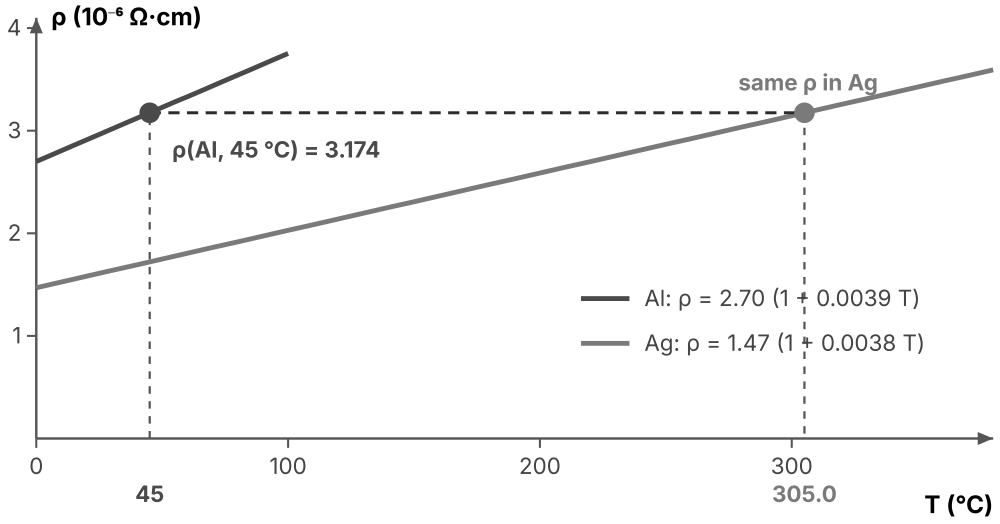


Figure 1. Problem 1: ρ = ρ0 (1 + αT) for Al and Ag; the Al wire at 45 °C has ρ = 3.174×10⁻⁶ Ω·cm and silver reaches it at 305.0 °C.

Strategy.

1. Convert the resistivities to SI.
2. Evaluate the resistivity of aluminium at 45 °C with the law of Slide 14.
3. Impose the same resistivity on silver and solve the linear equation for its temperature.
4. Check by substituting back, and compare with the solution that Slide 15 prints.

STEP-BY-STEP CALCULATION

1. Law of the thermal resistivity [Session 13 Slide 14]

The resistivity at temperature T (in °C, from 0 °C) of each wire is:

$$\rho_T = \rho_0 \left(1 + \alpha_T \, T\right)$$

2. Unit conversion to SI

One ohm-centimetre is one ohm times one hundredth of a metre:

$$1\ \Omega\,\mathrm{cm} = 1\ \Omega \times 10^{-2}\ \mathrm{m} = 10^{-2}\ \Omega\,\mathrm{m}$$

Applied to each resistivity at 0 °C:

$$\rho_{0,\mathrm{Al}} = 2.70 \times 10^{-6}\ \Omega\,\mathrm{cm} \times 10^{-2}\ \frac{\Omega\,\mathrm{m}}{\Omega\,\mathrm{cm}} = 2.70 \times 10^{-8}\ \Omega\,\mathrm{m}$$

$$\rho_{0,\mathrm{Ag}} = 1.47 \times 10^{-6}\ \Omega\,\mathrm{cm} \times 10^{-2}\ \frac{\Omega\,\mathrm{m}}{\Omega\,\mathrm{cm}} = 1.47 \times 10^{-8}\ \Omega\,\mathrm{m}$$

3. Resistivity of the aluminium wire at 45 °C

The product αT is dimensionless:

$$\alpha_{\mathrm{Al}}\, T_{\mathrm{Al}} = 0.0039\ ^{\circ}\mathrm{C}^{-1} \times 45\ ^{\circ}\mathrm{C} = 0.1755$$

$$1 + \alpha_{\mathrm{Al}} T_{\mathrm{Al}} = 1 + 0.1755 = 1.1755$$

$$\rho_{\mathrm{Al}}(45\ ^{\circ}\mathrm{C}) = 2.70 \times 10^{-8}\ \Omega\,\mathrm{m} \times 1.1755 = \mathbf{3.17385 \times 10^{-8}\ \Omega\,m}$$

4. Condition of the statement: the same resistivity in silver

$$\rho_{0,\mathrm{Ag}}\left(1 + \alpha_{\mathrm{Ag}}\, T_{\mathrm{Ag}}\right) = \rho_{\mathrm{Al}}(45\ ^{\circ}\mathrm{C})$$

Dividing both sides by $\rho_{0,\mathrm{Ag}}$, subtracting 1 and dividing by α_{Ag}:

$$1 + \alpha_{\mathrm{Ag}} T_{\mathrm{Ag}} = \frac{\rho_{\mathrm{Al}}(45\ ^\circ\mathrm{C})}{\rho_{0,\mathrm{Ag}}}$$

$$\alpha_{\mathrm{Ag}} T_{\mathrm{Ag}} = \frac{\rho_{\mathrm{Al}}(45\ ^\circ\mathrm{C})}{\rho_{0,\mathrm{Ag}}} - 1 = \frac{\rho_{\mathrm{Al}}(45\ ^\circ\mathrm{C}) - \rho_{0,\mathrm{Ag}}}{\rho_{0,\mathrm{Ag}}}$$

$$T_{\mathrm{Ag}} = \frac{\rho_{\mathrm{Al}}(45\ ^\circ\mathrm{C}) - \rho_{0,\mathrm{Ag}}}{\rho_{0,\mathrm{Ag}}\,\alpha_{\mathrm{Ag}}}$$

5. Numerical evaluation, one operation per block

Numerator, in ohm-metres:

$$\rho_{\mathrm{Al}}(45\ ^\circ\mathrm{C}) - \rho_{0,\mathrm{Ag}} = 3.17385 \times 10^{-8}\ \Omega\,\mathrm{m} - 1.47 \times 10^{-8}\ \Omega\,\mathrm{m} = 1.70385 \times 10^{-8}\ \Omega\,\mathrm{m}$$

Denominator, in ohm-metres per degree:

$$\rho_{0,\mathrm{Ag}}\,\alpha_{\mathrm{Ag}} = 1.47 \times 10^{-8}\ \Omega\,\mathrm{m} \times 0.0038\ ^\circ\mathrm{C}^{-1} = 5.586 \times 10^{-11}\ \frac{\Omega\,\mathrm{m}}{^\circ\mathrm{C}}$$

Quotient:

$$T_{\mathrm{Ag}} = \frac{1.70385 \times 10^{-8}\ \Omega\,\mathrm{m}}{5.586 \times 10^{-11}\ \Omega\,\mathrm{m}\,^\circ\mathrm{C}^{-1}} = \mathbf{305.0\ ^\circ C}$$

6. Substitution back into the law

$$\alpha_{\mathrm{Ag}} T_{\mathrm{Ag}} = 0.0038\ ^\circ\mathrm{C}^{-1} \times 305.02\ ^\circ\mathrm{C} = 1.15908$$

$$\rho_{\mathrm{Ag}}(305.02\ ^\circ\mathrm{C}) = 1.47 \times 10^{-8}\ \Omega\,\mathrm{m} \times (1 + 1.15908) = 1.47 \times 10^{-8}\ \Omega\,\mathrm{m} \times 2.15908 = 3.17385 \times 10^{-8}\ \Omega\,\mathrm{m}$$

7. The solution printed on Slide 15

Slide 15 rounds the aluminium resistivity to $3.173 \times 10^{-6}\ \Omega\cdot\mathrm{cm}$ before dividing:

$$T_{\mathrm{Ag}} = \frac{3.173 \times 10^{-6} - 1.47 \times 10^{-6}}{1.47 \times 10^{-6} \times 0.0038}\ ^{\circ}\mathrm{C} = \frac{1.703 \times 10^{-6}}{5.586 \times 10^{-9}}\ ^{\circ}\mathrm{C} = 304.87\ ^{\circ}\mathrm{C}$$

RESULT & CHECKS

Checks

- **Units of the coefficient term.**

$$[\alpha_T\, T] = {}^{\circ}\mathrm{C}^{-1} \cdot {}^{\circ}\mathrm{C} = 1$$

- **Units of the result.**

$$[T_{\mathrm{Ag}}] = \frac{\Omega\,\mathrm{m}}{\Omega\,\mathrm{m} \cdot {}^{\circ}\mathrm{C}^{-1}} = {}^{\circ}\mathrm{C}$$

- **Substitution.** Silver at 305.02 °C has 3.17385×10^{-8} Ω·m, equal to aluminium at 45 °C (step 6).
- **Rounding.** The 304.87 of Slide 15 differs from our 305.02 only because the slide rounds ρ_{Al} to four figures; our value keeps the exact product.

Interpretation

Silver conducts better than aluminium at the same temperature, so the temperature that equalises them is far above 45 °C: silver must be heated until its thermal resistivity has grown from its low value at 0 °C up to the aluminium value (step 5). The result is large because the thermal slope of silver, $\rho_{0,\mathrm{Ag}}\alpha_{T,\mathrm{Ag}}$, is small; as that slope tends to zero the temperature tends to infinity. The linear law is used up to 305 °C because the sheet fixes it; real metals bend away from a straight line at high temperature.

Check against the official solution

Item	Official	Computed	Comment
Temperature of the silver wire	305 °C	305.0 °C	agrees; Slide 15 prints 304.86 because it rounds ρ_{Al} to 3.173 (we get 304.87 with that rounding)